Yandel Polanco

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Education

- B.S. in Computer Science from University of North Carolina at Chapel Hill — Expected May 2028 | GPA 4.0
- A.A. in Computer Science from Miami Dade College — Graduated May 2026 | GPA 3.8

Work History

Teaching Assistant, Web Development at UNC CSSG — *Present*

- Teaching approximately 50 students in a weekly 2-hour web development course covering Git/GitHub, HTML, CSS, JavaScript/TypeScript, and React through live coding and hands-on assignments.
- Developing instructional materials and tutorials covering the setup of the development environment.
- Guiding students to showcase their projects, helping them plan and develop React applications that incorporate APIs, databases, or UI/UX concepts.

Game Developer at ROBLOX, Remote — *Feb 2023 to Present*

- Developed core gameplay mechanics in Lua alongside a 10-developer team for Literally FNF!, a Roblox game with 300 million total visits.
- Resolved outage-causing bugs by analyzing crash reports and deploying fixes; concurrent players recovered from approximately 200 to 1,000 within one week of the update.
- Composed original game music and refined tracks through playtesting and player feedback.

Founding Software Engineer at Nexus Horizon, Miami, FL — *Sept. 2025 to May 2026*

- Built a ROS 2 control system for a warehouse robotics prototype, translating velocity commands into coordinated control of 2 drive motors using Python, C++, and ESP32 firmware.
- Implemented encoder reporting at a target rate of 50 Hz using interrupt-driven firmware and developed wheel odometry to estimate robot position and orientation.
- Enabled LiDAR-based mapping and remote visualization by integrating RPLidar A1, SLAM Toolbox, and Foxglove on a NVIDIA Jetson Orin Nano and Raspberry Pi through ROS 2.
- Earned accelerator selection and won a startup competition; pitched robotics prototypes at eMerge Americas.

President & Hackathon Program Manager at INIT, Miami, FL — *Aug 2024 to May 2026*

- Developed technical workshops, networking events, and engagement initiatives, increasing chapter membership by 27 from an initial 5 members.
- Secured $30,000+ in external funding to sustain chapter operations and events.
- Led planning and execution of the SharkByte Hackathon with a 12-member organizing team, delivering a 36-hour event with 200+ attendees and 49 completed projects.
- Allocated $25,000+ in chapter funding and in-kind support through partnerships with Microsoft, Cybriant, Red Bull, React Miami, and Miami Dade College to fund SharkByte.

Undergraduate Researcher at Miami Dade College, Miami, FL — *Feb. 2025 to Aug. 2025*

- Built a full-stack website for literary analysis using Flask, Python, HTML/CSS w/ Bootstrap, and JavaScript.
- Implemented user authentication, database connectivity, and file uploads for secure user interactions and document management.
- Created an interface for machine learning models to analyze novels and short stories through scalable backend processing pipelines.

Artificial Intelligence Extern at Outamation, Remote — *Feb. 2025 to March 2025*

- Classified, extracted, and analyzed mortgage documents and resumes in Google Colab using Python, Fitz, Tesseract OCR, PIL, and JSON to process and structure document data.

Projects

Portfolio Website yandel.dev — *2026*

- Built a personal portfolio website using vanilla HTML, CSS, and JavaScript to showcase projects.

Velocity yan-dev.itch.io/velocity — *2024*

- Delivered a complete movement-focused FPS within a 6-month deadline, leading a 3-person team and implementing all gameplay programming in Unity and C#.
- Produced the game's 3D assets in Blender and original soundtrack in FL Studio, owning development across code, art, and audio.

Technical Skills

Languages: Java, JavaScript, TypeScript, Python, Lua, C#, C++, HTML/CSS
Frameworks & Runtimes: React.js, Node.js, Flask, ROS 2
AI/ML & Libraries: NumPy, Hugging Face, LayoutLM, Meta Llama API, Tesseract OCR
Developer Tools: Git, GitHub, PostgreSQL, Arduino IDE, Visual Studio, Unity, Blender, FL Studio